

LEVELING UP DURING LOCKDOWN: THE SURGE OF SOCIAL GAMING IN COLLEGE
STUDENTS DURING COVID-19 AND ITS POST-PANDEMIC IMPLICATIONS

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Abstract

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The number of Multiplayer Online Video Games (also known as MPOGs) players steeply increased during the COVID-19 pandemic. When lockdowns and social isolation restrictions were set in place many sought opportunities to turn their once in-person social interactions into meaningful virtual ones. The research presented will address the question: What factors enhance social connection and build virtual communities in MPOG players? Two research methods have been used, focusing on three popular video games played during the COVID-19 pandemic. The three games focused on are *Minecraft*, *Among Us*, and *Animal Crossing: New Horizons* to evaluate the effectiveness of communication tools used to help cope with social isolation. Ethnographic research has been conducted using Discord, a popular online communication platform popular with gamers, to explore interactions in MPOG communities and identify factors that keep users engaged. Interviews of Gen Z gamers at the University of California, Berkeley help identify features that promote competence, autonomy, and relatedness in MPOG players. It has been found that across all three games issues such as in-game chat limitations and negative interactions with other players pose issues in creating a positive social environment. Thus the research conducted leads to creating a framework that is aimed at solving the issues found by using human-centered design principles to tackle commonly missed issues that arise when developing MPOGs. With an impactful framework game developers can utilize it to design video games that foster a sense of mastery, control, and connection in virtual gaming communities.

Keywords: COVID-19 • Social Isolation • Human-Centered Design • User experience • Multiplayer Online Games • Video Games

Contents

Abstract	i
Contents	ii
Introduction and Background	1
Literature Review	4
How the Pandemic Has Shaped Gen Z	4
Terms “Gaming” & “Gamer” Over the Years	5
Quarantine Gaming	7
Negative Effects	11
Methodology	13
Ethnographic Research	13
Minecraft	14
Among Us	15
Animal Crossing	16
Interviews	16
Results	18
Ethnographic Research	18
Minecraft	18
Among Us	19
Animal Crossing	20
Interviews	20
Playing MPOGs	21
COVID-19’s Impact	23
Negative Game Experiences	25
Coding Research	26
Framework	28
Conclusion	30
Limitations & Challenges of Current Research	30
Possible Future Research	31
Bibliography	33

Introduction and Background

For many the COVID-19 pandemic affected people's ability to connect with others. Many members of Gen-Z were impacted by the sudden transition from in-person experiences to fully remote, losing connections with friends and family during an important time of their lives necessary to their developing brains (Harari et al., 2022). Members of Gen Z require in-person social interactions as they get into the later stages of their brain development to help them build important social connections (Harari et al., 2022). Social connections that are built in these stages of life are often still important as they get older. Older Gen-Z members, specifically those who are currently in college, around their last teen years to their early twenties, are just entering the beginning stages of joining the workforce. These members of Gen Z are the most impacted by the effects that the COVID-19 pandemic has caused on their social lives. Their transition into undergraduate or graduate school experiences and learning how to adapt to virtual job opportunities require Gen Z to utilize their social connections. For college students, these social connections have been limited due to the inability to be a part of recruitment events and to get to know classmates or colleagues. Strong social connections can help create friendships, mentors, and good workplace relationships.

The COVID-19 pandemic also caused severe social isolation for every generation. Not being able to see loved ones and family for months or years can take a toll on many people's mental health. To help cope with social isolation, many had to find new hobbies or interests and be a part of virtual communities. Many members of Gen Z play video games of some kind and play more than any generations before them (Khan, 2023). For many, video games can be a form of escape. Video game sales grew by 73% in April 2020 compared to the previous year (Richter, 2020). Many games, specifically Multiplayer Online Games, known as MPOGs for short, saw a

quick rise in popularity during the pandemic. MPOGs such as *Minecraft*, *Among Us*, and *Animal Crossing* all have benefited from the pandemic however, some have been around longer than others. During this thesis, I researched these three popular video games to understand how they became successful during the pandemic and how Gen Z utilized them in order to gain social connections.

Minecraft, created in 2009, has always been popular since its release. *Minecraft* is a video game that appeals to different types of gamers of various ages and skill levels. *Minecraft*'s structure as a game mashes up more survival-based games, such as *Call of Duty*, and more casual games that allow for creative building environments, such as *The Sims*, into one. Players can choose their difficulty for the game or mode meaning they can choose a more building mode game or a survival mode depending on what type of game they'd like to play that day. While *Minecraft* is a widely well-known video game the number of active users has died down throughout the years (Curry, 2023b). When the pandemic hit, however, the game saw a jump in monthly active users from 91 million in 2019 to 131 million in 2020, when pandemic shutdowns were in full swing (Curry, 2023b). *Among Us*, which was released in 2018 and was relatively new when the pandemic first began (Curry, 2023a). The game saw a huge spike from half a million in 2019 to 300 million monthly users in 2020 (Curry, 2023a). During the COVID-19 lockdowns in 2020, Nintendo released the 5th video game installment of their *Animal Crossing* series. *Animal Crossing: New Horizons* is a game that allows players to build their island and interact with their friends. The game has a cutesy art style and characters and a more relaxed game-style setting compared to *Minecraft* or *Among Us* which was able to allow a wider range of possible game players and age ranges, specifically those not interested in stressful games that rely on bullets or puzzles. This meant people who typically are not considered gamers, meaning

those who may play multiple video games of a certain genre, could learn how to play a video game in a low-stakes environment and could get their foot through the door. *Animal Crossing: New Horizons* completely has the pandemic to thank for its enormous success. While *Animal Crossing* installments have been around long before the pandemic, social distancing drew a new crowd of existing gamers and would-be gamers as the game does require experience in the previous installments and also is easy to learn for the first time. *Animal Crossing: New Horizons* also changed how a video game could be accessed which changed the definition of a gamer in some ways. Handheld games were pretty popular pre-COVID-19 such as playing games on the Nintendo DS and even Nintendo changed how generations could access gaming through the WiiU, however, neither Nintendo product had the amount of success the Nintendo's newest product had during the pandemic. Nintendo's Switch, which was released in 2017, brought in a whole new audience of gamers. The Switch allowed for portable playing but also allowed players to hook up their switch to a TV, it also had controllers that could be used by one person or two people making the console easy to share on the go. Being a gamer no longer required expensive gaming machines such as a PC or consoles like PlayStation or Xbox.

Some have found the term "gamer" embarrassing or a term that needs to be reevaluated to make gaming culture less toxic and often stereotyped in harmful ways (Alexander, 2014). Others have found that the term gaming has changed during the 21st century due to new technology. While I believe that gaming after the pandemic needs to be refined and separated from previously defined ideas of gaming from the 21st century pre-pandemic. Using my completed research, I developed a framework that can be used by game developers and companies to evaluate if their multiplayer games can help build virtual communities and a positive environment that supports the mental health of all players, especially Gen Z.

Literature Review

How the Pandemic Has Shaped Gen Z

Gen Z can be defined as anyone born between 1997 and the 2010s meaning they were primarily mid-teens to young adults during the COVID-19 pandemic. Members of Gen Z grew up fully immersed in technology, with social media and digital pop culture being an influence throughout their lives. Due to their highly digital lives, however, members of Gen Z also suffer from high amounts of anxiety and depression (Harari et al., 2022). In addition to growing up under very different circumstances such as economic factors and digital media compared to other generations, the pandemic also halted their transition from high school to college to the workforce during a time when they are still developing their social network (Harari et al., 2022). Dr. Tali Te'eni Harari and her team in Israel researched the different effects that the pandemic had on both Gen Z and Gen X. Dr. Harari and her team looked at the differences between the two generations with how they coped as well as adapted during the COVID-19 pandemic. With Gen Z's need to quickly adapt to Zoom and online learning portals for school, virtual interviews for job recruitment, and their first experiences with the workforce being a work-from-home situation, this generation was seen to be more resilient and more open to combining new technologies with things they were familiar with in order to have a better and healthier quarantine experience (Harari et al., 2022). Gen Z was able to utilize video game spaces and other digital modes that they were already familiar with as a way to turn in-person activities such as doing workplace icebreakers, high school and college graduations, picnics, and even college tours into virtual ones. An example of a once in-person turned digital is UC Berkeley's college tours on their very own UC "Blockeley" where students could take virtual tours of a to-scale model of campus made in *Minecraft*. Simulation and multiplayer online video games were the

closest many could get to salvaging and recreating what were supposed to be the fun years of their teen and young adult years that they would not get back.

Terms “Gaming” & “Gamer” Over the Years

The 21st century saw a change in how video games were developed and played, redefining what is considered a video game. Furthermore, the COVID-19 pandemic has fully shifted who can be considered a “gamer” today. There was a shift in games focusing on narrative and stories over tactics as well as video games becoming considered somewhat of an art form (Jagoda, 2017). A notable game achieving this was the game *Journey* from 2012 by the studio Thatgamecompany. *Journey* focuses on beautiful and aesthetic graphics with a simpler and interpretive story over complex controls and tactics to complete the game (Miller, 2015). The game won several awards and was even the first video game ever to be nominated for a Grammy due to its phenomenal soundtrack (Pinchefsky, 2012). *Journey* is a game that randomly pairs players with strangers to have them attempt to reach the end of the game. Players are required to do so without being able to chat or text each other, something not common in multiplayer video games. I picked up the game during the pandemic and found myself creating wordless connections with strangers. I was also completely in awe of the graphics and peacefulness of the game. There are so few games I have encountered that were able to achieve a peaceful environment while still being a multiplayer game, which often can get out of hand due to bot spamming and immature gamers. While there were games of a similar style that existed before *Journey*, they did not all gain traction as they hadn’t created a successful environment to create an anonymous connection with others. After *Journey*’s release, there have been more and more games that learned from its success and have gained traction. These games focus on simplicity but there have been several debates these art-styled games can even be considered games if they

don't focus on tactics (Jagoda, 2017). Patrick Jagoda, a media theory and game studies professor at the University of Chicago, writes in his paper "Videogame Criticism and Games in the Twenty-First Century" about the shift in game development and some of the newest genres in the industry. Jagoda mentions about MPOGs as well as esports and the community surrounding them in his paper. Esports has often been highly criticized as a real sport due to its virtual nature. During the pandemic, however, when quarantine requirements restricted team sports and other close contact activities, esports allowed people to play virtual versions of their favorite sports, these games can even be streamed just like one might watch an NBA tournament (Smith, 2020). Esports was one of the few televised live competitions that did not have to be put on hold due to the pandemic, and in fact, benefited from it as it allowed people to discover new ways to play competitively.

Leigh Alexander's article "Gamers' don't have to be your audience. 'Gamers' are over." published in Game Developer was an interesting article to engage with. It can be used to see how much the game community has shifted from its original assumption back in the arcade or Gameboy days. At first glance, one might mistake Alexander for critiquing the gaming community negatively and being unnecessarily harsh when she begins the article by describing a familiar stereotype people might recognize if they've ever been to a game convention. Alexander (2014) describes game culture as an embarrassing subject filled with a feeling of ick when one thinks of a "gamer", someone who is socially withdrawn, specifically men. However, Alexander (2014) turns the conversation onto the game developers and marketers themselves, questioning them for taking advantage of this community. She states that game marketers have caused the decrease in diversity in gaming due to the primary target they have chosen. These games tend to exploit the mental health of socially withdrawn individuals for the profit of their own company

by creating female characters with bigger busts and other methods of appealing to men while selling them an unrealistic and unattainable lifestyle (Alexander, 2014).

I however also found that the audience for video games shifted during COVID-19, as games such as *Animal Crossing: New Horizon* are marketed to a very different audience than “gamer” stereotypes. The game seems to be aimed toward a younger audience however has a player based in many age groups. While the article by Alexander was not entirely in the direction of my study, which focuses on college students and MPOGs I found this critique helpful in understanding how one might develop a game to help reduce two types of social isolation, quarantine isolation and social anxiety. If a developer focuses on developing for a group affected by only one form of isolation, it might make the other form of isolation more prominent which could cause unintended negative outcomes. A video game “solving” quarantine isolation could lead to gamers reducing their social circles as they may prefer connections with strangers over friends in their community.

It is important to note, however, that since 2014 when Alexander first wrote her critique the gaming community and its developers have changed especially during the COVID-19 Pandemic. Competitive and esports players have also increased since 2017 when Jagoda first wrote his paper on the shift of gaming in the 21st century. There has been a large shift, economically, in the 21st century. Due to this shift, it might be time to see how much gaming has changed, specifically in how it engages with its community.

Quarantine Gaming

The shift in gaming can be seen in studies about gaming during the COVID-19 Pandemic. J. P. Bergeron wrote “Happy Home Quarantine: A Content Analysis of Reddit Posts Demonstrating Gratifications of Life-Sim Gaming during COVID-19” as their thesis dissertation

for the University of Nebraska in 2022. Bergeson (2022) focuses on trying to understand what outcomes players receive when playing games like *Animal Crossing: New Horizons* and if it helps with their mental health. “Happy Home Quarantine” uses posts on the social media forum Reddit as a form of research to understand how players utilize the game. Bergeson (2022) sorts through the most upvoted posts in r/AnimalCrossing so that they can look for community engagement and posts that “resonates with the highest number of r/AnimalCrossing visitors” (p. 14). 3

Bergeson’s (2022) thesis led me to research more about the effects that quarantine had on many. The dissertation does not have a clear audience it is focusing on for its research other than players of a certain game, thus I wanted to know how much quarantine had an impact on Gen Z. The thesis focuses heavily on *Animal Crossing: New Horizons* for its research. This research will expand on Bergeson’s research in order to learn more about other types of games and their differences from *Animal Crossing*. Bergeson’s methodology only focuses on one kind of research, observing and analyzing Reddit posts from the *Animal Crossing* forum. This form of research seems to lack analyzing deeper human interactions as posts are completely anonymous, which dissociates a user from their emotions and personality. These posts are not fully written to engage in conversation, but rather to share options and experiences. Reddit posts are also mostly written and edited in a style that is meant for a post to gain traction in order for it to be seen so conversations may only be about majority opinions.

A study conducted by Ryuichi T. Kishimoto, Yogi Tri Prasetyo, Satria Fadil Persada, and A. A. N. Perwira Redi was one of the few I was able to find that focused on one game’s effect on just members of Gen Z during the COVID-19 Pandemic. The study focuses on ratings of satisfaction for mobile gameplay while also bringing up an important discussion on how peer

pressure may affect gaming (Kishimoto et al., 2021). Many MPOGs require groups of players to play together to complete tasks, while these games can be played with strangers through public servers, it may be out of many player's comfort zones to be on voice chats with people they do not know at all. Many members of Gen Z are also underaged meaning that parents need to limit interactions with strangers as they can be inappropriate and harmful. Both the issues of comfortability with strangers one might find themselves buying different game consoles or games to be able to play with people in their close social circles especially if those social circles are already involved in a video game community. For example, one might buy the Nintendo Switch as well as the cartridge for *Animal Crossing: New Horizons* just because their friends are already building islands together. Many college campus events and clubs had students download *Among Us* for free to play during icebreaker sessions. These already existing video game communities from close social circles somewhat create peer pressure to download video games to keep these social circles thriving during quarantine isolation. While the idea of peer pressure may seem negative, data showed that those who downloaded a game, because their friends told them about it, had a big impact on the satisfaction of gameplay as they were able to feel closer to their peers (Kishimoto et al., 2021). The study does, however, address its limitations as its study group was only based in the Philippines and only focused on one of the country's most popular games. My research focuses on primarily the effect COVID-19 had on Gen Z in the United States of America as my ethnographic research has been conducted with North American players and my interviews take place with UC Berkeley students. However, these interviews come from students with a variety of backgrounds such as international students and minority backgrounds in gaming to understand how gaming was utilized by college students as a whole.

There are many other takes on whether video games helped or had an effect on people's

mental health during the pandemic. Games can help stimulate cognition and socialization, ease depression and anxiety, and overall provide a feeling of escapism during quarantine (Barr & Copeland-Stewart, 2022). Matthew Barr and Alicia Copeland-Stewart (2022) conducted a study to learn how games were utilized during the pandemic. They had survey participants answer questions about games they played and if they had negative or positive experiences with playing those games. 46.5% of their participants were students meaning a large portion of their survey was taken by Gen Z however it also focuses on those employed, retired, or consider themselves homemakers (Barr & Copeland-Stewart, 2022). After finalizing synthesizing and their research, Barr and Copeland-Stewart defined seven positive ways that gaming helped with well-being during COVID-19.

Mental Health was the first defined effect. Games like *Animal Crossing* were highlighted by survey participants as they found their favorite games or certain games in particular genres to help them cope and overall made them feel less anxious (Barr & Copeland-Stewart, 2022). Games also were a distraction from social media as time spent scrolling through numerous posts that might be anxiety-provoking was instead spent on playing video games (Barr & Copeland-Stewart, 2022). The second defined effect was Stress Relief which is closely related to mental health, as players can unwind by playing a fun game of their choosing (Barr & Copeland-Stewart, 2022). Players who are stressed can take a break and put their focus on games that require creativity or problem-solving. The next two effects personally felt intertwined with each other and the key to a good video game, Escape and Agency. Players can choose who they want to play as and which virtual world they'd like to be "transported" to. One could decide to do a stealth mission on a military base or be a nomad wandering through beautiful landscapes. Those choices can create a "distraction" making players feel immersed in the world they chose to

surround themselves with and not the realities of the COVID-19 pandemic (Barr & Copeland-Stewart, 2022). The next effect was Cognitive Stimulation as it keeps the brain occupied but still running (Barr & Copeland-Stewart, 2022). Many games have puzzles or tasks to get to various points in the game which gives the brain some exercise (Barr & Copeland-Stewart, 2022). The sixth defined effect was Normalisation. During a turbulent time, players were able to fall into a routine they felt comfortable with to replace those they could no longer do due to quarantine (Barr & Copeland-Stewart, 2022). The last defined effect was Socialization, whether it was for keeping in touch with friends during isolation or bonding more with those who were living in a bubble together. “The social capital [...] accumulated through online multiplayer gaming, for example, has been associated with improved well-being, health, and happiness” (Barr & Copeland-Stewart, 2022). Barr and Copeland-Stewart’s study was incredibly helpful in getting a better understanding of the field of study. It also narrowed down which parts the research should focus on more that Barr and Copeland-Stewart may have missed, as the study does not focus on a particular age range, but still has a high amount of Gen Z participants.

Negative Effects

While there are multiple discussions previously mentioned that state the positive impacts that gaming can have on Gen Z during the pandemic, there are other arguments that counter this. Some of the minor concerns that appear when looking into pandemic gaming, such as how players may find themselves using gaming as a distraction or keep them from completing more important tasks such as school or work (Barr & Copeland-Stewart, 2022). While this issue is not a large concern, as some players found themselves focusing on the positives from the gaming experience, it is still something to consider as players should still be able to complete personal

important tasks outside of gaming (Barr & Copeland-Stewart, 2022).

Gamers with problematic coping mechanisms may be at harm as while the feeling of escape may be good for some gamers it can lead to dissociation for others (Pallavicini et al., 2022). *Hikikomori*, a Japanese term used to describe extreme social withdrawal, has been on the rise again since COVID-19. Pre-pandemic there was already a discussion on how video games were causing young adults to self-isolate, a study looked into Australian and North American young adults who played multiplayer games to see how socially withdrawn they are (Stavropoulos et al., 2019). Many players already may have a skewed perception of reality when they engage in avatar building as characters can be built on how one either perceives themselves to be or how they wish to be, allowing them to escape from reality (Stavropoulos et al., 2019). The number of hours gaming can also affect one's mental health as a gamer can spiral into an addiction to gaming, specifically, those who play over 8 hours a day will tend to pick gaming over any other activity (Stavropoulos et al., 2019). It's important to understand this huge negative impact that gaming has had on some as many game developers are either unaware of the effects their games may have on individuals or may be exploiting these individuals for profit.

Understanding the positives as well as the negatives of multiplayer games is necessary for understanding some of the patterns that appear in problematic games to reduce the risk of depression and anxiety from social isolation. Games such as *Minecraft* or *Animal Crossing* can escalate quickly on how much of a positive or negative effect it can have on one's mental health, but these also depend on the person who is playing them and the structure of the video game. Video games should allow for a seamless transition from virtual to in-person social interactions when gamers come out of quarantine requirements.

Methodology

Expanding on Bergeron's research for their dissertation, this thesis will focus on two ways of understanding how Gen Z uses video games in order to combat social isolation and create meaningful virtual interactions. It will also attempt to find out the inequities the current gaming community creates. There are a wide range of generations that play *Animal Crossing: New Horizons* however they all may use it for different purposes specific to their age, such as mocking up classrooms, graduations, weddings, and even workplace socials (Bergeron, 2022). By looking at the broader audience of players who play *Animal Crossing* and solely gathering research from online forums it's harder to analyze which tools were beneficial for those suffering the most from the pandemic. This thesis comprises of two research methods: ethnographic research and interviews. I analyzed conversations on a social media platform, Discord as well as focused on talking to gamers themselves through interviews. Ethnographic research will be conducted on Discord to see live conversations of game interactions rather than asynchronous discussions on a forum such as Reddit. Interviews will be narrowed to a specific age group of Gen Z college students.

Ethnographic Research

Ethnography in a virtual space (also known as Netnography) is a method of analyzing virtual communications, interactions, and experiences that adapts the in-person observation methods of anthropology to the study of digital cultures. Fieldwork is conducted in the virtual space of three public Discord servers. Discord is a social media platform that provides text and voice chat tools. It is a platform popular with video gamers that allows users to engage in asynchronous and synchronous informal conversations. The ethnography research will be conducted in the official Discord servers for three MPOGs – *Animal Crossing*, *Among Us*, and

Minecraft. The rationale for selecting these three MPOGs is because of their steep increase in popularity globally during the COVID-19 pandemic as mentioned earlier.

I chose to only research three games as each game was made during different times of gaming culture some before and some during the pandemic. *Minecraft* was created way before the pandemic began and has always been a popular multiplayer game but saw an increase in playtime during the pandemic like UC Berkeley's "Blockeley" mentioned before however some of the player bases are still regular players since its initial release. Since *Among Us* was released right before the pandemic and *Animal Crossing: New Horizons* was released right during the pandemic, their player base was primarily players affected by the pandemic seeking virtual communities. Researching more than three would have led to the focus of the paper being disjointed and disorganized.

For the ethnography research, notes would be collected of observed Discord interactions within the three sites. These notes would include in-game conversations and even unrelated conversations that happened on these servers to understand how the platform was being used to aid in-game experiences. I spent 4 hours each day for 4 weeks collecting data on unique interactions and compiling them into 3 detailed field notes per day. Field notes consisted of the number of participants who took part in the interaction, the behavior of the interaction (such as friendly conversations or aggressive behavior), and any important quotes. By the end of 4 weeks, I had collected over 60 interactions.

Minecraft

Minecraft's official Discord server is set up very differently than most official game servers. The space allows for Mojang employees who helped develop *Minecraft* to interact with players and the server's moderators who oversee if users are following server rules. What is also

different about this server compared to others is how organized but also how restrictive the space is due to how many members there are. The server only displays moderators and staff as members on the right-hand chart in Discord, compared to other very large servers that list every member. Listing all members in a server can get glitchy if the server is incredibly large and also removes privacy for those a part of the server as strangers can interact with one's Discord profile. *Minecraft* limits who is seen to show the most important people, moderators, and employees. By only removing casual gamers from the list, it somewhat removes a sense of community for those wishing to seek a community with other players on the server who are okay with having their profile public. The server also enforces a feature called "slow mode", a feature that can only be set up by moderators in order to limit how many messages can be sent in a certain period. Slow mode is enabled to reduce spamming, an issue seen too often in these kinds of spaces. While slow mode does not eliminate spamming it reduces the time one can send either repetitive or hateful messages giving time for moderators to intervene. For the *Minecraft* server, the slow mode feature is set to one message only every twenty seconds.

Among Us

Among Us' Discord platform is set up to include spaces for game chat, off-topic conversations, and multiple voice chat rooms for players to communicate with each other during game sessions. The space offers more synchronous conversations compared to other official Discord game channels. Staff and employees are also included in this server however they do not engage with the channel's users as much as those from the *Minecraft* server. The *Among Us* server has set rules but does not enforce them through channel slow mode but instead by allowing players and users to take part in vote-kicks which allow people to collectively decide if one needs to be eliminated from a game rather than an individual, such as a moderator, deciding

for themselves. While the server has many asynchronous or text-related conversations in channels, more immersive conversations occur in voice channels. The channels are set up in lounges for casual conversations, more fun-related activities such as karaoke for bonding events, voice channels for other games such as *Minecraft*, *Codewords*, or even Chess, and forty-five channels made for in-game team conversations. These forty-five voice channels can be used by those seeking others for a game round in *Among Us*.

Animal Crossing

The Discord server for *Animal Crossing* is newer compared to the other two servers as *Animal Crossing* was released right during the pandemic, so the server is more geared towards bringing the community together. The server is not oriented towards voice channels but rather having players engage in text conversations together to share their experience in the game so far and trade proposals. For *Animal Crossing* many players talk about needing to get up at odd hours to avoid their character “voiding” or crops within the game spoiling. *Animal Crossing* requires players to be on top of the game even after hours. The style of the game has players engaging constantly as the world follows the player’s own time from sunrise, sunset, and even nighttime. When a player opens the game during the night, the environment in the game reflects nighttime. In *Animal Crossing*, farms, and characters are needed to be taken care of on a timely basis similar to houseplants or pets.

Interviews

To learn more about Gen Z’s gaming habits and the impact that COVID-19 had on them I conducted interviews with college students who have played at least one of the three ethnographic study games. Eight interviews were conducted with University of California undergraduates. Interview participants were found through gaming clubs on campus, game

development courses, and the e-sport center. Through interviews, the goal is to learn more about participants' perspectives on playing MPOGs and engaging in virtual gaming communities through 30-minute interviews. Interview questions help form a story about the interview participant's experiences with gaming during pre-pandemic, during the pandemic, and post-pandemic. Getting an overall story of the interview participant's experience during the pandemic helps form an understanding of their relationship with gaming and trying to find social interactions in a virtual world.

Results

Ethnographic Research

Minecraft

Most interactions that I logged from the *Minecraft* Discord server were neutral. There were lots of forms of taunting but also basic conversations. Many players complained about the slow mode when it would increase to a minute instead of twenty seconds. Others would try to have off-topic discussions with one another about other games or events in their own lives. Moderators would often intervene to try to get conversations to solely be about the game itself rather than about debates that would sometimes arise.

Taunting in the space often included somewhat harmless remarks about other games, such as another game called *Roblox* being “better”, telling others to “touch grass”, or that someone was acting “cringe”. While these taunts are not severe hate comments they can still take away from conversations about the game and make players feel bad about themselves. Taunting in these servers is often immature and is also ignored most times. The activity within the *Minecraft* server was not as active as the other two servers I chose to research later on. This low activity usage is because the game had been out many years before the pandemic and its audience has slowly died down since its release. Even though the game has seen an increase in users from the pandemic, most players seem to have used *Minecraft* to communicate with existing social circles meaning that using a server with a broader audience is not necessary.

Based on the servers’ users’ reactions, it's apparent that the server is somewhat poorly run and does not engage in more social aspects. For a game such as *Minecraft*, Discord servers, for particular in-game worlds created by users, *Minecraft* servers, would probably be more active for

more social interaction both synchronously and asynchronously. The game itself also has a text chat within the game so using a space such as Discord is not fully necessary.

Among Us

Interactions within this Discord space were more off-topic conversations but allowed players to bond with one another. Some interactions in the general chat were users playing mini guessing games together built into the server such as guessing country flags and doing math problems quickly. Players utilize spaces to play similar games outside of the space as well such as online web games, the server acts as a more multi-use platform which eliminates the need for joining other Discord servers and gives the sense of a community. Many moderators and members of the server will work together to make the place safer, an example of a sticky situation and unfortunately frequent issue in the *Among Us* server is spamming as there is no chat slow mode. One of the interactions logged was a new user joining the server only to quickly spam the chat with hate comments in the general chat space. Users actively tried to alert moderators by pinging them to remove the spam comments and ban unwanted users.

The space was primarily used however for the voice chat feature to allow people wanting to find a group to play with to find spots to fill. The voice chat was more active than the text chat features and used by more people. In the text chat, there were often users who were frequent visitors and would chat together around the same time of the day each day. During observations, they would come back the next day around the same time frame to see how everyone was doing and start new conversations about unrelated topics. Some of these users became somewhat “server celebrities” as most people in the server would know them because of how much they engaged in conversations with others. These conversations in the text chat channels were used more to create a community rather than to just play *Among Us*. Those who just wanted to play

Among Us without interacting with the broader community could just do so by being a part of voice chat.

Animal Crossing

Animal Crossing participants like to replicate their real world within the game, such as their gaming setups or bedrooms, many interactions involve players replicating their lives in a virtual space which helped the game thrive during the pandemic. A virtual life allowed players to share personal experiences and socialize while following COVID-19 safety guidelines.

Interactions also had users discussing events that were held on certain islands to promote wellbeing, or related to pride and heritage months such as Disability Pride Month.

Some of the notes taken were how many times users would reference other games to point out similarities or differences. One interaction noted was a discussion on *Minecraft* on how time and sun/moon rises/sets worked on the plane. They discussed how *Minecraft's* length for a day was much shorter than real life. In this discussion, they asked questions on how time worked in *Animal Crossing* and how it is based in real-time meaning a day in the game is the same as a day in the player's world. Unlike *Minecraft* which's hours are sped up which makes days shorter, *Animal Crossing* follows the user's actual time so certain tasks need to be done at certain times of the game. This interactive feature is also an engaging part of the game which keeps players coming back and playing the game throughout the day. Due to time being a necessity in the game, this also created conversations on Discord where players shared tips on when to access certain critters.

Interviews

Interviews were the key part of this research study as talking to those directly affected by the COVID-19 pandemic and using MPOGs to improve conditions of social isolation. Eight

interviews were conducted from September 2023 to November 2023. These interviews were done with various students of different academic backgrounds throughout the University of California. Findings from the interviews were divided into three key sections, how players used MPOGs, how players felt about isolation during COVID-19, and key negative game experiences.

Playing MPOGs

Some MPOG players started playing games during the pandemic while others grew up surrounded by media such as the Nintendo DS or web games. An interview participant told me about her experience playing educational web games when she was younger and how that inspired her to continue playing similar games with her friends. Regardless of experience in gaming or when one started playing, the main reason for many joining the MPOG community was to play with their friends or family. Another experience shared was a student who grew up playing games with her two brothers on their Nintendo Wii and now together playing on the Nintendo Switch. Another student also shared her experience bonding with her brother by playing *Halo* on Xbox. Games create a bonding experience with one another. Another participant talked about his experience collecting physical characters for the game *Skylanders* to play with his friends during elementary school. In *Skylanders*, each character had certain powers that could be collected to play by putting characters on a “portal” which displayed chosen ones within the game. The student said that these game characters were popular birthday gifts and he’d sometimes even trade them with friends if he had duplicates or wanted a better character he knew a friend had. The social aspect of gaming could be seen as fun, competitive, and elevated already shared in-person experiences.

From interviews, experiences playing MPOGs were mostly positive, as their interactions with the games we discussed made them want to return to the game in the future. Some even

played the game multiple times a week, even today. All the interview participants mentioned how they used MPOGs to unwind. They found some of these games to be relaxing and a great escape from work or school pressures. While many MPOGs can be played by oneself, most interview participants shared a similar experience of preferring to play games with their close friends or family members rather than with strangers. From discussions with interview participants who played with strangers over people in their close social circle, it seemed as though these interactions were more inconsistent depending on who they were randomly paired up with or decided to join in a server. These interactions were dicy as give or take someone could be paired with someone who was either not good at the game or someone mean.

Playing with friends also allowed for conversations to occur outside of in-person experiences as after classes or school one could play games at home separately but still together within the game. Other positive experiences were finding new communities with similar interests as well as meeting new people at gaming events on campus. While some preferred using MPOGs to strengthen their already created and maintained in-person friendships, others used MPOGs to find and meet new friends. MPOG clubs on the University of California, Berkeley's campus allowed people to meet their new potential friends in person and put game user names to faces. Many clubs meet once a week or every other week on campus and play games together and discuss strategies on how to beat different levels or how to tackle a raid. One interview participant said this experience of being in a campus club and meeting people with a mutual interest that was not academically related was necessary for them to maintain a healthy education/life balance. This need was even further necessary due to social isolation during the pandemic which hampered their ability to be on campus their first year of college and to be involved in college campus clubs whether professional or informal-social.

Pre-pandemic modes of communication while playing MPOGs remained either in-person or through in-game chat features. Discord was used as a replacement for voice chat in games that did not offer mic or chat conversations with one another. Players often use these tools to communicate when they decide to meet up to play and which segments of a game to play. An interview participant said that they joined a Discord server that was sent to them by a friend, they did not know any other people on the server but quickly became friends with them and the medium of a text chat or voice chat that did not show their face helped them overcome social anxiety slowly.

COVID-19's Impact

When asked about how many hours they spent gaming pre-COVID versus during COVID-19, all interview participants agreed that their time spent gaming during the COVID-19 pandemic increased greatly compared to before. Some interview participants mentioned trying to quickly find ways to interact with their friends while still social distancing. Some found themselves stuck playing games for hours being addicted to playing MPOGs. Luckily from interview participants, there were no signs of dangerous social withdrawal behaviors as they still engaged in social events from their school or with family. However, that does not mean that these behaviors do not exist or have not affected Gen Z or college in any way. There may be students on campus who do feel socially withdrawn and prefer using video games as a form of communication over in-person ones.

For many of the interview participants, however, now that social isolation restrictions have been lifted, they no longer play video games as much as they used to. Also with the pressures of college, it is harder to find time to play games when they are busy studying for exams. Regardless, most interview participants mentioned that they still found that they

benefited from using MPOGs and video games, in general. It can be stated that they overall found MPOGs to improve their mental health during the COVID-19 pandemic.

Some participants said they played party games with each other during the pandemic through Discord. These games kept them engaged with each other and helped them check up on each other each week. A participant told me about how he and his friends played online party games once a week with his group of friends from high school and they would often set up days to play an MPOG shooter together to complete tasks within it and get in-game currency. One participant told me about how their club on campus used *Among Us* as an icebreaker during their first week of getting to know each other. Playing a video game allowed the space to define itself as informal and for strangers to loosen up a bit. Another interview participant mentioned that she and her teammates from one of her classes decided to meet up on *Animal Crossing* at the end of her first virtual semester as a way to celebrate the end of their final project. She said that they were not close during the class itself when they were working on the project but found using *Animal Crossing* as a great way to get to know each other in an informal environment other than Zoom.

Another interesting detail gathered from interviews was how a club hosted their end-of-semester banquet as well as even hosting an alumni event in a game similar to *Animal Crossing*. They ordered or made food at home instead of meeting up at a restaurant or on campus but called each other in a virtual game with a voice chat feature. The game had puzzle games within it and spaces where small clumps of people from the large club gathering could talk to each other if within a certain distance of each other. They said this game was the closest they had seen to something that replicated an in-person large gathering as the game muted people outside of a certain distance from them allowing for individual conversations or smaller group

conversations. Using a platform such as this made virtual meetings more enjoyable rather than just simple video-style conferences.

Negative Game Experiences

For some interview participants, some issues barred them from fully enjoying the MPOG community or particular games. One participant told about how she found strangers she played with in *Minecraft* would try to make sexual gestures within the game because they knew she was a girl based on her feminine in-game avatar. She said these interactions would only happen when they were near her and would make her incredibly uncomfortable. However, she said she still kept coming back to the game as these experiences were not often. Overall she said that her interactions in the game were overall more positive but that was only guaranteed if she played with friends she knew instead of strangers. Unfortunately, this was not the only comment about gross sexualized experiences that were shared in interviews. One participant shared with me briefly how many characters in a game they have played for a while have slowly become sexualized which has changed the demographics of the players vastly. She has slowly seen there be a largely male population playing in the game. Another participant stated a very similar experience in a similar game saying that the group they are in that meets up to play has slowly become all-male. In these certain gaming communities, women have become a minority and often feel uncomfortable participating.

More experiences include fear of joining voice channels, issues with the fan base, and releases of the game becoming inaccessible. Many of the existing issues such as the one above plus the ones previously mentioned mostly affect women and those a part of the queer community. While voice channels allow players to engage with the game much more closely with others, they also provide a possibly toxic and unsafe environment for women and the trans

community. One interview participant mentioned how she does not join voice chats in games ever, even though it would greatly improve her team's coordination and sense of direction during a match of an MPOG shooter game. Another interview participant, who stated she is trans, said she avoids what she knows will be toxic MPOGs, such as Valorant, altogether.

Lastly many stated how overwhelming and tiring it gets to join so many different gaming communities such as being a part of multiple Discord channels. Some prefer only to put their efforts into one Discord channel and use it for multiple points of interaction such as playing other games that aren't the server's main intentions. It allows people to interact with a similar community many times and to become familiar with them rather than playing with a new group of people only for *Among Us*, a different group for *Minecraft*, and a completely separate group for *Animal Crossing*. Multiple ongoing conversations can become overwhelming over time.

Coding Research

When conducting research was finally complete I moved on to coding my research. I picked out key similarities from my research and organized them into categories, tagging them to my notes. The codes are as follows:

Moderator interactions are seen when moderators would intervene to remove users from Discord or remove chats. These showed up somewhat frequently on certain channels such as the *Among Us* Discord which has a habit of getting spammed frequently due to the type of demographic that plays the game, younger audiences, and because the server has limited restrictions that bar these issues from happening. This is directly tied with the second code which was on hate comments and spamming, this was discussed in both ethnographic research and interviews. Certain interactions with people caused many players and even interview participants to feel uncomfortable when playing certain games or has changed how they interact with the

space whether it's changing their gaming style or avoiding ways to interact with others socially within the game. This issue is a key point I will be focusing on in the framework I have developed.

Within a specific gaming community, many still utilize the space to interact with other games. This led me to develop a specific code as I wanted to highlight how a single space can become used for many different interactions and allows gamers to find a community in one space rather than across many different Discord servers which can become overwhelming according to some interview participants. However, through my ethnographic research, I found that some Discord servers, such as the *Minecraft* server, were strict on not having off-topic conversations and only provided a space for *Minecraft* players. A desire to find community was a need I heard and saw over and over again. Even before the COVID-19 pandemic, people joined MPOGs to create a community and meet people with similar interests, they also used the space to further strengthen their in-person friendships or play with family members. Lastly, micro-communities and minority populations were a code that was evident after digging through research as I found that some communities tried to host events for Pride Month to make the space accessible to minority communities and give a sense of openness.

Framework

With the research found from this thesis, I worked on creating a framework or series of considerations that can be used by game developers and studios to assess how effective their game is in providing resources for those experiencing isolation. The framework uses principles from human-centered design to consider the players and users first. Human-centered design principles often require developers of a product to do extensive research on their audience and test their proposed product with their intended audience to receive feedback and iterate on the product making it stronger. Many times these processes are overlooked despite how simple they may seem. User testing only occurs to find and fix bugs within a game as well as to find out if players find a game too easy or too hard to play. User testing in most game development studios, however, does not focus on selecting a diverse amount of participants from various backgrounds to share their experiences playing in the MPOG environment created. This framework intends to create a solution to this problem and allow game studios to talk directly with their player base and even with those they may have not considered as potential gamers.

Before officially launching a video game I believe it is important and best that developers test their game material with a diverse group of participants. These participants must reflect the actual percentage of gamers currently playing games in the same genre of multiplayer games as well as include participants from underrepresented communities such as other genders and ethnicities as well as other minority groups. This may include women and people from the non-binary community, people with low-vision or hearing disabilities, and players from Black and indigenous backgrounds. Game marketers must also consider the price they decide to propose their game to consider accessibility and avoid excluding players from multiplayer games that may help benefit them.

Before game development, game developers should ask and consider statements from participants on their experiences playing other MPOGs from similar genres. Having participants create a pros and cons chart could be incredibly helpful in identifying key features to consider and avoid as participants from minority communities could have feedback that may often be overlooked or not seen during development. Developers should implement at least two key concerns that fit within the reason of each minority community when creating their game. These implementations can range from creating options for color-blind players, AI that scans for hate comments, or safe spaces and communities within the game for players of minority groups such as random pairing with only LGBTQ+ members in a game like *Among Us* to create a safe space for them.

When completing user testing on early versions of the game, participants should also be asked to reflect on their experience interacting with other players within the game as well as give one piece of feedback that allows for improvement. The game should also consider how their tools for communication will be used and if they are effective in creating positive spaces in their game. Questions for participants in order to understand if their game fulfills this requirement can be “If you were to play this game what additional tools would you use to play this game?”, “Who are you most likely to play this game with?”, and “What interactions would you most likely wish to avoid in this game?”. These sorts of questions can help developers and game designers consider what tools to implement and which tools to moderate for a safe online experience. The game should be iterated to implement feedback. User testing should happen again to see if these concerns have been solved and if no new concerns arise.

Conclusion

Limitations & Challenges of Current Research

There are some limitations that arose during the research process of this thesis. One includes the amount of participants gathered for interviews. The initial goal of the amount interviews to be conducted was around ten to fifteen participants. Only eight interviews were completed. Finding interview participants who played at least one of the three games for the study and were also UC Berkeley students was not too hard to do due to my method of gathering them. What was a concern was while I did manage to receive messages from people willing to participate the population of participants was not diverse. I received a highly male population willing to participate in general multiplayer games and almost only female audience for interviews on *Animal Crossing*. I wanted my research to cover topics that minority gamers faced when playing MPOGs to help grow their community. While I was not able to complete ten to fifteen interviews I made sure that my interview population was as diverse as I possibly could. Interview participants came from different ethnic and racial backgrounds, income, and sexual or gender orientation. I however was not able to find participants with disabilities that may hinder their ability to play certain multiplayer games.

Another limitation was the inability to narrow my ethnographic research to only North American participants. While I chose to use the North American and English-speaking servers for each game's Discord all participants are anonymous meaning they may or not be located in North America. Due to this, I chose to conduct research only during the daytime to make sure to filter out users that may be from out of the States.

Lastly, a limitation I had was getting in contact with game development studios. It would have been helpful more me to get to sit down with game developers and learn how I can create a

framework that would be useful for them. Unfortunately, I did not have the capacity during this project to do this process.

Possible Future Research

There is still much to do when it comes to making sure MPOGs are accessible and considering the mental health of players who may be struggling with social isolation. Now that the world has resumed to almost fully in-person experiences this research may not seem as necessary or prevalent anymore, however, it can still be used for two major reasons. One is to make sure that in case of a future global pandemic, there are already existing practices in place that can be used to help promote better mental health. These tools must be as accessible as possible to minimize the effects social isolation may have for those facing future possible pandemics or who are required to stay indoors due to illness. The other important reason is that this research helped us understand what other communities may have experienced social isolation or negative MPOG experiences before the COVID-19 pandemic. These issues have always existed however they were never fully researched until they happened to a larger population of people. Research about gaming associated with social isolation only increased during the pandemic however social isolation has always been an issue, especially in cases such as *Hikikomori* players who are extremely socially withdrawn causing massive issues for their mental health and even their own families who they may be detached from.

In the future, I hope to continue this research to help minimize some of the issues video games may have caused for those who are vulnerable and affected by social isolation. Further research would include continuing to increase interview participants and finding more participants from diverse backgrounds such as those with disabilities. I also hope that I will be able to get in contact with game developers and work on iterating and finalizing my framework

so that it can begin to have a positive impact on the gaming community. By using the human-centered design approach when working on my framework I can see how developers actively use the framework or help with user testing to see if games would improve with it. As I work towards finalizing my framework I can turn it into a handbook or worksheet that can be easily accessed by game developers wishing to consider inclusivity in their game.

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